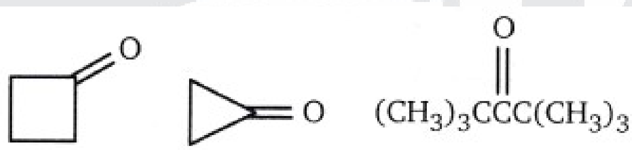
CHEMISTRY TARGET : JEE Advanced - 2023	**CAPS – 19** **Aldehydes, Ketones & Carboxylic Acid**

1. Compound (X) $C_9H_{10}O$ gives yellow coloured ppt with 2,4 DNP but does not give red coloured ppt with Fehling's solution. (X) on treatment with NH_2OH/H^+ gives compound (Y) $C_9H_{11}NO$. (Y) when treated with PCl_5 gives isomeric compound (Z). (Z) on hydrolysis gives propanoic acid and aniline. What will be the correct structure of (X), (Y) and (Z) ?

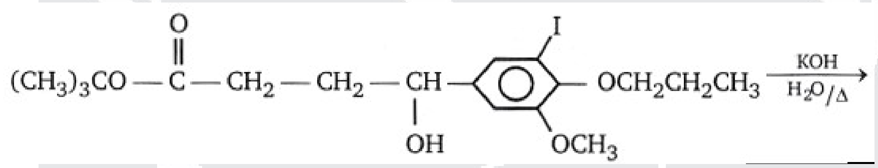
	(X)	(Y)	(Z)
(A)	$C_6H_5-\underset{\underset{O}{\parallel}}{C}-C_2H_5$	$C_6H_5-\underset{\underset{HO-N}{\parallel}}{C}-C_2H_5$	$C_2H_5-\underset{\underset{O}{\parallel}}{C}-NH-C_6H_5$
(B)	$C_6H_5-\underset{\underset{O}{\parallel}}{C}-C_2H_5$	$C_6H_5-\underset{\underset{N-OH}{\parallel}}{C}-C_2H_5$	$C_2H_5-\underset{\underset{O}{\parallel}}{C}-NH-C_6H_5$
(C)	$C_6H_5-\underset{\underset{O}{\parallel}}{C}-CH_3$	$C_6H_5-CH_2-\underset{\underset{N-OH}{\parallel}}{C}-CH_3$	$CH_3-\underset{\underset{O}{\parallel}}{C}-CH_2-NH-C_6H_5$
(D)	$C_6H_5-\underset{\underset{O}{\parallel}}{C}-C_2H_5$	$C_6H_5-\underset{\underset{N-OH}{\parallel}}{C}-C_2H_5$	$C_6H_5-\underset{\underset{O}{\parallel}}{C}-NH-C_2H_5$

2. Rank the following in order increasing of the equilibrium constant for hydration, $K_{hyd.}$ (smallest value first).



1 2 3

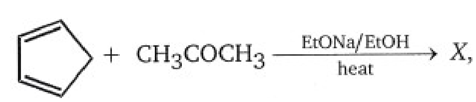
(A) 1 < 2 < 2 (B) 3 < 1 < 2 (C) 2 < 1 < 3 (D) 2 < 3 < 1

3. 

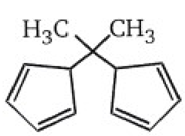
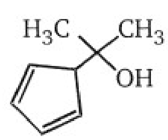
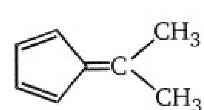
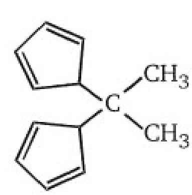
Total number of products obtained in above reaction is:

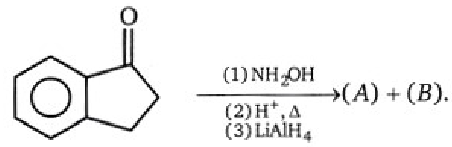
(A) 2 (B) 3 (C) 4 (D) 5

4. In the reaction

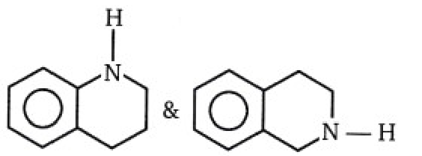


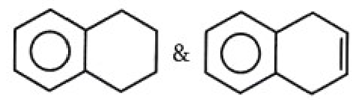
the product (X) is

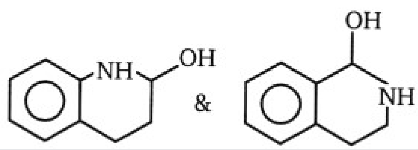
(A)  (B)  (C)  (D) 

5. 

Product (A) & (B) are

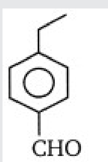
(A) 

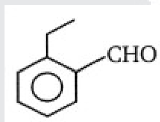
(B) 

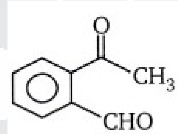
(C) 

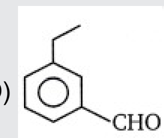
(D) 

6. An organic compound with the molecular formula $C_9H_{10}O$ forms a 2,4-DNP derivative, reduces Tollen's reagent and undergoes Cannizaro reaction, on vigorous oxidation it gives 1,2-benzenedicarboxylic acid. Structure of organic compound is:

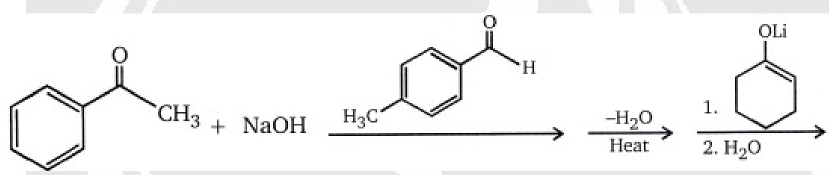
(A) 

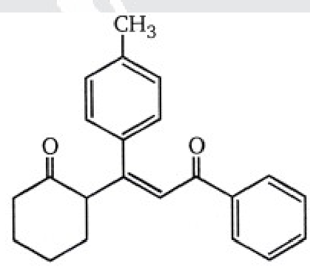
(B) 

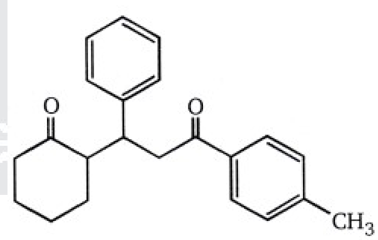
(C) 

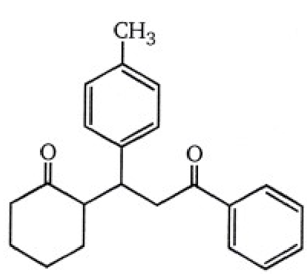
(D) 

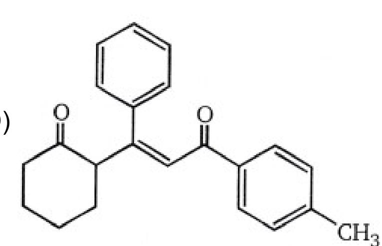
7. Predict the major product of the following reaction sequence

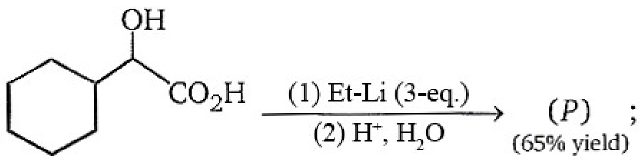


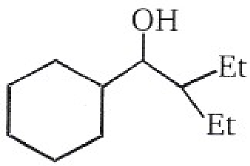
(A) 

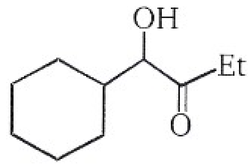
(B) 

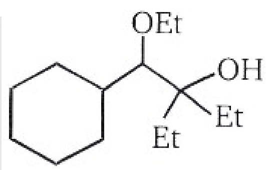
(C) 

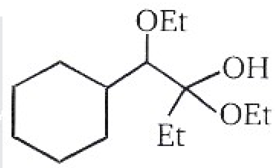
(D) 

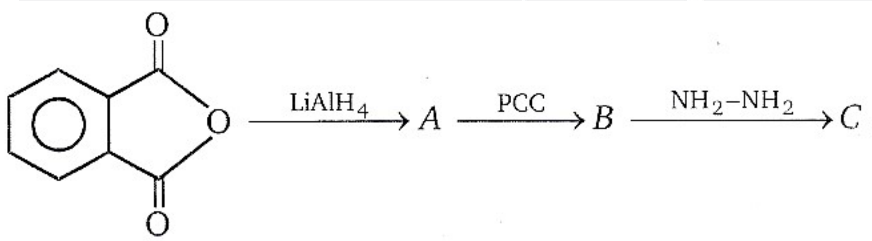
8.  ; Product (P) of the reaction is :

(A) 

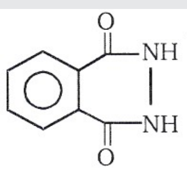
(B) 

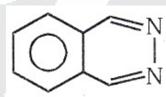
(C) 

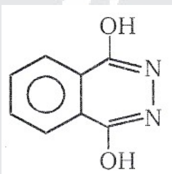
(D) 

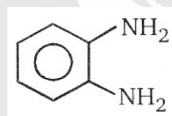
9. 

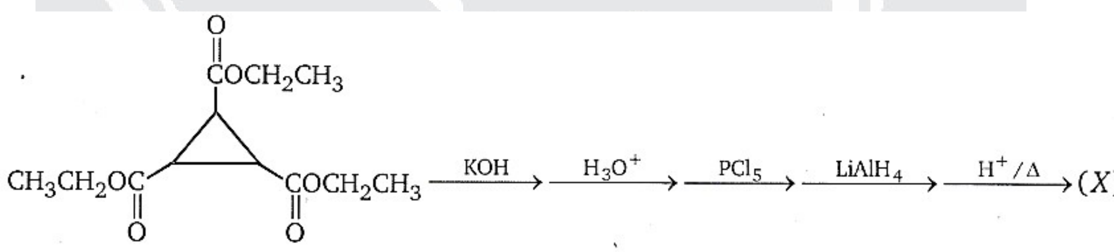
Product (c) is

(A) 

(B) 

(C) 

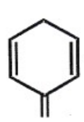
(D) 

10. 

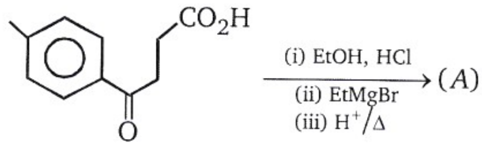
Product (X) is

(A) 

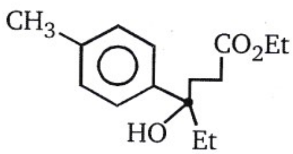
(B) 

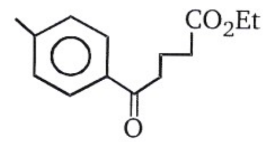
(C) 

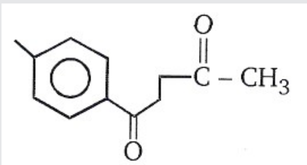
(D) 

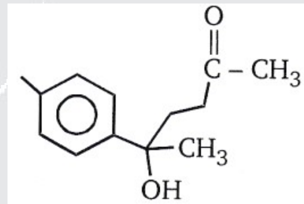
11. 

Product (A) of the reaction is

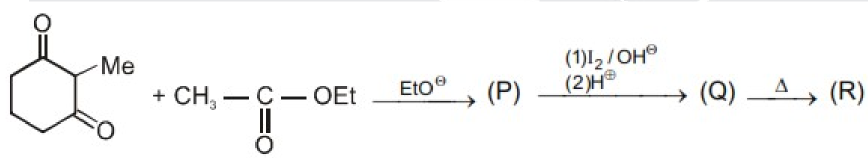
(A) 

(B) 

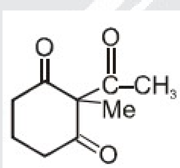
(C) 

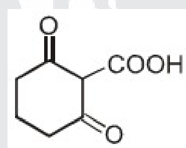
(D) 

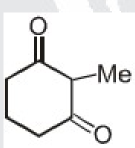
MCQ

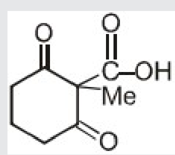
12. 

The products of the above reaction is / are :

(A) 

(B) 

(C) 

(D) 

13. The following conversion is/are possible by

$$Ph-CH_2-CH=O \longrightarrow \underset{\text{(Phenyl alanine)}}{Ph-CH_2-\underset{\displaystyle NH_2}{\underset{|}{CH}}-COOH}$$

(A) $\xrightarrow{KCN\ /\ H_2O} \xrightarrow{NH_3,\ \Delta} \xrightarrow{H_3O^{\oplus},\ \Delta}$

(B) $\xrightarrow{KCN\ /\ NH_4Cl} \xrightarrow{H_3O^{\oplus},\ \Delta}$

(C) $\xrightarrow{HCN\ /\ NaOH} \xrightarrow{SOCl_2} \xrightarrow{NH_3} \xrightarrow{H_3O^{\oplus},\ \Delta}$

(D) $\xrightarrow{Br_2\ /\ CH_3COOH} \xrightarrow{NH_3} \xrightarrow{CrO_3\ /\ H^{\oplus}}$

14. Identify product in the following reaction sequence

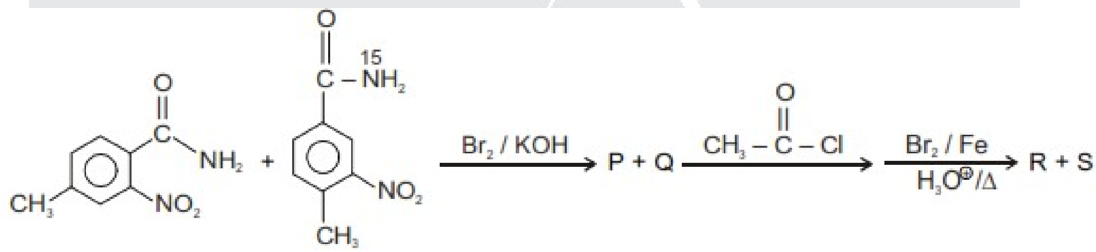
(A) $C_6H_5-\overset{\overset{\displaystyle O}{\|}}{C}-OH$

(B) $C_6H_5-\underset{\underset{\displaystyle O}{\|}}{C}-CD_3$

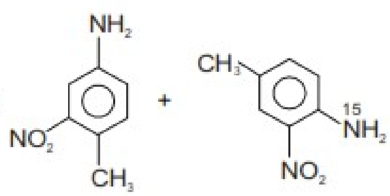
(C) $C_6H_5-\underset{\underset{\displaystyle O}{\|}}{C}-Cl$

(D) $C_6H_5-CH_2-\underset{\underset{\displaystyle O}{\|}}{C}-Cl$

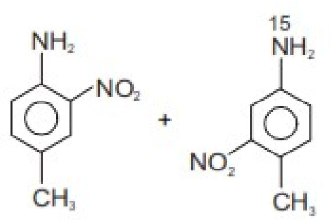
15. The correct option for products P, Q and R, S in the following sequence of reaction is / are :



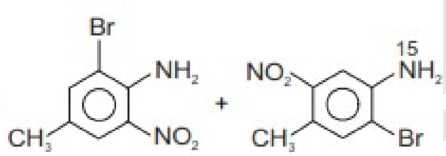
(A) P & Q are



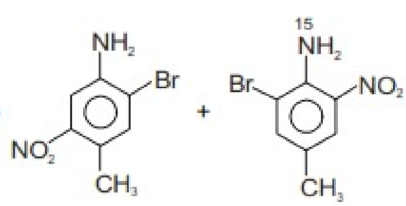
(B) P & Q are



(C) R & S are



(D) R & S are



Comprehension for Question 16 to 18

Observe the following sequence of reaction and answer the questions based on it

$$\text{Phenylacetylene} \xrightarrow[-\,CH_4]{CH_3MgBr} x \xrightarrow[\text{ii) } H^{\oplus}]{\text{i) } CO_2} y \xrightarrow[HgSO_4]{H_2O\,/H_2SO_4} z \xrightarrow{\Delta} w$$

16. Compound z is:

(A) $Ph-CH_2-\overset{\overset{\displaystyle O}{\|}}{C}-COOH$

(B) $Ph-CH<\begin{matrix}COOH\\COOH\end{matrix}$

(C) $Ph-\overset{\overset{\displaystyle O}{\|}}{C}-CH_2-COOH$

(D) Ph – CH_2 – COOH

17. Which of the following statement is not correct

(A) y decolourises Br_2 /H_2O solution

(B) z on heating liberates CO_2 gas

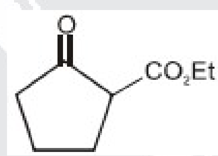
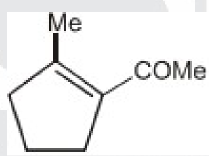
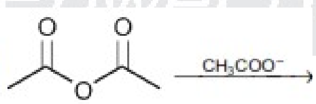
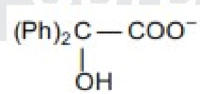
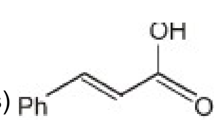
(C) w on reaction with NaOI gives yellow ppt

(D) x liberates H_2 gas with Na metal

18. Which of the following compound give benzoic acid on $KMnO_4$ oxidation

(A) w (B) y (C) z (D) all.

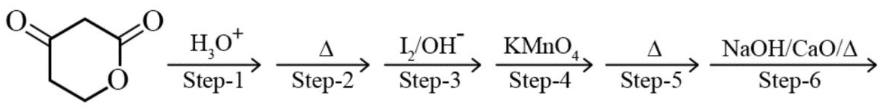
19. Match the column

Column I	**Column II**
(A) $MeCO(CH_2)_4COMe \xrightarrow[\text{(ii) } H^+ / \Delta]{\text{(i) } OH^{\ominus}}$	(p) 
(B) $(CH_2)_4<\begin{matrix}CO_2Et\\CO_2Et\end{matrix} \xrightarrow{\overset{\ominus}{O}Et}$	(q) 
(C) PhCHO + 	(r) 
(D) $PhCOCOPh \xrightarrow{OH^{\ominus}}$	(s) 

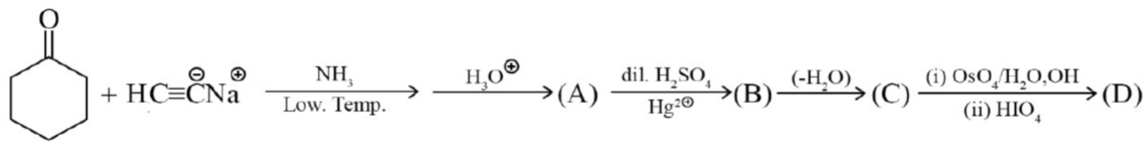
20. Complete the following sequence of reactions :

$$\begin{matrix}H_3C\\H_3C\end{matrix}>CH-C<\begin{matrix}O\\NH_2\end{matrix} \xrightarrow[\Delta]{P_2O_5} (A) \xrightarrow{H_3O^+} (B) \xrightarrow{Br_2/P} (C) \xrightarrow{AgOH} (D) \xrightarrow[-2H_2O]{\Delta} \text{Diester.}$$

21. In how many steps decarboxylation reaction is taking place.



22. Complete the following reactions:



Write the structures of A to D and give the IUPAC name of (D).

23. When acetone is treated with excess of benzaldehyde in the presence of base, the crossed condensation add two equivalents of benzaldehyde and expels two equivalent of water and forms [X]. Identify the structure of [X] when [X] reacts with NH_2OH how many stereoisomers are formed.